

BREMEN, Germany

WMO: 102240

Lat: **53.0450N**Lon: **8.7978E**Elev: **17**StdP: **14.69**Time Zone: **1.00 (EUC)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.1	19.7	10.0	9.2	19.0	14.1	11.2	21.5	31.0	44.8	28.4	45.0	8.8	80	0.593

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.8	85.7	67.4	81.2	66.2	77.8	64.6	69.4	81.0	67.7	78.3	66.1	75.2	9.5	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.9	95.5	73.7	64.2	89.9	71.6	62.5	84.8	70.3	33.4	81.3	32.0	78.4	30.8	75.3	75.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.5	21.2	18.9	DB	11.1	91.7	5.8	3.6	6.9	94.3	3.5	96.4	0.3	98.4	-3.9	101.0	(4)
(5)			WB	10.4	72.9	5.7	1.3	6.3	73.8	2.9	74.5	-0.3	75.2	-4.5	76.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.8	35.1	36.3	41.2	49.1	55.1	60.8	65.4	64.2	58.1	50.8	42.7	37.8	(6)	
	DBStd	12.55	8.12	7.45	6.91	7.04	6.90	6.14	6.11	5.53	5.26	6.21	6.70	8.18	(7)	
	HDD50	1938	462	385	280	102	25	1	0	0	3	66	232	382	(8)	
	HDD65	5745	927	804	737	479	316	156	71	83	217	441	669	845	(9)	
	CDD50	1862	1	1	7	74	183	326	478	441	246	90	13	2	(10)	
	CDD65	192	0	0	0	0	10	31	83	58	10	0	0	0	(11)	
	CDH74	1772	0	0	0	14	101	291	801	483	78	4	0	0	(12)	
	CDH80	493	0	0	0	0	15	62	266	138	12	0	0	0	(13)	
(14) Wind	WSAvg	9.4	11.2	10.7	10.8	9.4	9.0	8.6	8.4	8.1	8.4	8.8	9.3	10.4	(14)	
(15) Precipitation	PrecAvg	28.0	2.0	1.7	1.6	2.2	2.7	2.4	3.2	3.2	2.2	2.3	2.4	2.5	(15)	
	PrecMax	34.6	3.5	3.7	2.4	4.1	4.9	5.9	5.9	6.6	7.2	5.9	3.7	5.1	(16)	
	PrecMin	22.3	0.5	0.5	0.7	0.5	0.8	0.9	1.3	0.6	0.6	0.9	0.6	0.4	(17)	
	PrecStd	4.1	1.0	1.0	0.6	1.1	1.1	1.5	1.1	1.7	1.4	1.3	0.9	1.5	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.0	55.5	64.6	76.6	81.0	85.9	91.7	89.4	81.0	71.9	60.7	55.7	(19)	
		MCWB	50.6	48.6	53.6	58.9	65.4	67.9	69.0	68.5	66.3	62.2	56.4	52.4	(20)	
	2%	DB	51.4	51.6	59.0	70.8	76.7	81.0	86.6	84.0	75.5	66.2	57.2	53.4	(21)	
		MCWB	48.7	47.5	50.7	56.1	62.0	66.6	67.4	66.7	64.5	60.0	54.6	50.6	(22)	
	5%	DB	48.2	49.0	54.7	66.3	72.8	77.1	82.4	79.0	71.2	62.6	55.2	50.9	(23)	
		MCWB	46.5	46.1	49.0	54.9	59.8	64.7	66.8	65.1	62.1	58.1	52.5	48.7	(24)	
	10%	DB	46.0	46.5	51.7	62.0	68.1	73.1	78.4	75.3	67.6	59.8	52.1	48.2	(25)	
		MCWB	44.5	44.1	47.1	53.1	58.0	62.5	65.4	63.8	60.3	56.3	49.3	46.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.7	50.3	55.2	60.3	67.5	70.7	71.7	71.1	68.3	63.1	57.7	53.1	(27)	
		MCDB	53.9	52.2	62.4	70.3	79.5	81.7	83.6	83.1	78.8	70.0	59.6	55.3	(28)	
	2%	WB	49.1	48.2	52.0	57.9	64.2	67.9	69.7	68.5	65.1	60.6	54.9	51.2	(29)	
		MCDB	50.8	50.8	56.6	67.6	73.7	78.6	82.4	79.1	73.3	65.3	56.9	53.0	(30)	
	5%	WB	46.2	46.1	49.9	55.9	61.5	65.6	68.0	66.7	63.0	58.7	52.5	48.9	(31)	
		MCDB	47.7	48.7	54.1	64.6	69.4	74.0	79.3	76.0	69.5	62.1	54.9	50.5	(32)	
	10%	WB	43.9	44.0	47.6	53.7	58.8	63.6	66.4	65.0	61.1	56.7	50.1	46.2	(33)	
		MCDB	45.3	46.1	51.3	61.0	66.2	70.9	76.2	73.1	65.9	59.8	51.8	47.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.1	9.7	13.4	17.2	17.4	17.3	17.8	17.7	15.7	13.2	9.1	7.9	(35)	
		MCDBR	9.3	12.0	19.5	23.9	23.5	23.7	24.5	23.9	21.4	16.6	10.6	8.8	(36)	
	5% WB	MCWBR	8.7	9.5	12.7	12.8	12.0	11.9	10.0	10.3	11.5	11.1	8.7	8.1	(37)	
		MCWBR	8.7	11.1	17.2	21.1	20.8	21.2	21.9	19.6	18.3	14.8	10.5	8.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.327	0.346	0.378	0.413	0.411	0.409	0.425	0.417	0.390	0.362	0.341	0.319	(40)	
		taud	2.327	2.322	2.248	2.201	2.246	2.273	2.243	2.280	2.344	2.395	2.356	2.317	(41)	
	Ebn at Noon		204	238	253	259	266	267	260	255	247	229	196	183	(42)	
		Edh at Noon	19	26	34	41	41	40	41	37	31	24	19	16	(43)	
(44) All-Sky Solar Radiation	RadAvg		190	406	763	1261	1562	1648	1576	1314	954	539	236	146	(44)	
	RadStd		20	56	82	152	170	128	209	97	115	55	25	12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	+2.07	+3.20	N/A	N/A	N/A	N/A	-272	N/A	N/A	
(47)	Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-239	N/A	N/A	(47)

Nomenclature: See separate page